SIMATS ENGINEERING

Department of Computer Science & Engineering

ASSESSMENT TOOL 2-COMPARATIVE ANALYSIS

Course Code:	**CSA12**	**Course Title:**	**Computer Architecture for Multicore Systems**
CO Assessed:	**CO4**	**Assessment:**	**ASSESSMENT TOOL2-COMPARATIVE ANALYSIS**
Weightage:	**30%**	**Total Marks:**	**25**
CO4 Statement:	*Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.*		
Student Name:		**Reg.No.:**	
Department:		**Date:**	

ANNEXURE A
COMPARATIVE ANALYSIS

1.Compare L1, L2, and L3 cache levels in terms of latency, bandwidth, and throughput, and analyze their impact on processor performance.
2.Differentiate SRAM and DRAM by evaluating their latency, bandwidth, and throughput characteristics in cache and main memory design.
3.Compare virtual memory with paging and cache memory mechanisms in terms of access latency and system throughput.
4.Analyze the differences between NAND Flash and DRAM with respect to latency, bandwidth, and suitability for hierarchical memory systems.
5.Compare MRAM and RRAM technologies by examining their performance in terms of access time, throughput, and energy efficiency.

Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

MARKS ALLOCATION

	Scope of Items (20%)	Comparison Depth (25%)	Use of Evidence (20%)	Critical Thinking (20%)	Clarity of Presentation (15%)
Q1					
Q2					
Q3					
Q4					
Q5					

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. L1 vs L2 vs L3 Cache

Feature	L1 Cache	L2 Cache	L3 Cache
Location	Inside CPU core	Near/inside CPU core	Usually shared among cores
Capacity	Smallest	Medium	Largest
Latency	Lowest	Low	Higher than L1/L2
Bandwidth	Highest	High	Lower than L1/L2
Throughput	Very high	High	Moderate–high
Typical use	Most frequently used data/instructions	Frequently used data	Shared data between cores
Cost per bit	Highest	High	Lower

Analysis

- **L1** provides the fastest access and highest bandwidth, so a high L1 hit rate directly improves CPU performance.
- **L2** is larger but slightly slower. It reduces the number of accesses to L3 and main memory.
- **L3** is much larger and usually shared. It improves multicore performance by allowing cores to share frequently used data.

Performance order:

L1 → L2 → L3 → DRAM

As cache level increases, capacity increases but latency generally increases.

Conclusion: A well-designed cache hierarchy reduces average memory access time and prevents the CPU from frequently waiting for slower DRAM.

2. SRAM vs DRAM

Feature	SRAM	DRAM
Storage cell	Flip-flop	Capacitor + transistor
Latency	Very low	Higher
Bandwidth	Very high	High
Throughput	Very high for small working sets	High for large data transfers
Refresh required	No	Yes
Density	Low	High
Cost	Expensive	Cheaper
Main application	CPU cache	Main memory

Analysis

SRAM:

- Does not require refresh.
- Has very low access latency.
- Can provide extremely high bandwidth.
- Requires more transistors per bit, making it expensive and less dense.
- Therefore, it is ideal for L1, L2 and parts of L3 cache.

DRAM:

- Stores data using capacitors and requires periodic refreshing.
- Has higher latency than SRAM.
- Provides much greater capacity at lower cost.
- Suitable for main memory.

Conclusion

SRAM → Cache memory → Low latency/high speed

DRAM → Main memory → High capacity

Using SRAM for cache and DRAM for main memory provides a good balance between speed, capacity and cost**.**

3. Virtual Memory/Paging vs Cache Memory

Feature	Cache Memory	Virtual Memory with Paging
Purpose	Speed up memory access	Provide larger logical memory space
Storage	SRAM	DRAM + secondary storage
Access latency	Very low	Higher
Transfer unit	Cache line	Page
Speed	Extremely fast	Much slower during page faults
Throughput	High	Lower when frequent paging occurs
Managed by	Hardware/cache controller	OS + MMU
Main problem	Cache miss	Page fault

Cache mechanism

The CPU first searches the cache. If the required data is present, it is a cache hit, resulting in very low latency.

CPU → Cache → DRAM

Paging mechanism

Virtual memory divides memory into pages. If a required page is not in RAM, a page fault occurs and the operating system must retrieve it from secondary storage.

CPU → MMU → RAM → SSD/HDD if page fault

Analysis

Cache memory improves processor throughput because the CPU can access frequently used data quickly. Paging improves the apparent memory capacity but can significantly reduce throughput when page faults occur.

Conclusion: Cache is primarily a performance mechanism, while virtual memory/paging is primarily a capacity and memory-management mechanism.

4. NAND Flash vs DRAM

Feature	NAND Flash	DRAM
Type	Non-volatile	Volatile
Latency	Much higher	Much lower
Bandwidth	Moderate to high depending on interface	High
Throughput	Good for bulk storage	Very high for active workloads
Persistence	Retains data without power	Loses data without power
Capacity	Very high	High
Refresh	Not required	Required
Main use	SSD/storage	Main memory

Analysis

DRAM:

- Provides low-latency access compared with NAND Flash.
- Supports high-bandwidth random and sequential memory operations.
- Ideal for active programs, operating systems and working data.

NAND Flash:

- Has much higher access latency.
- Provides high density and non-volatile storage.
- Excellent for storing operating systems, applications, files and AI models.
- Modern SSDs use controllers and parallel NAND channels to achieve high throughput.

Hierarchical design

CPU → Cache → DRAM → NAND Flash

Frequently used data stays in cache/DRAM, while large persistent data remains in NAND Flash.

Conclusion: DRAM is suitable for fast working memory, whereas NAND Flash is suitable for large, persistent storage.

5. MRAM vs RRAM

Feature	MRAM	RRAM
Technology	Magnetic storage	Resistive switching
Volatility	Non-volatile	Non-volatile
Access time	Very fast	Fast
Read latency	Low	Low
Write performance	Fast	Fast, technology-dependent
Throughput	High	High potential
Energy efficiency	High	Potentially very high
Endurance	Very high	High, but device-dependent
Main advantage	Fast, durable non-volatile memory	High density and low-power potential

MRAM

- Stores data using magnetic states.
- Retains information without power.
- Provides fast read/write access.
- Has high endurance.
- Can be used where fast non-volatile memory is required.

RRAM

- Stores information by changing the resistance of a material.
- Non-volatile.
- Has the potential for high density and low-energy operation.
- Can be useful for embedded memory and AI/neuromorphic applications.
- Actual performance varies significantly with the RRAM implementation.

Performance comparison

Access time:
MRAM and RRAM can both provide fast access, with the exact result depending on the device architecture.

Throughput:
Both can support high throughput, particularly with parallel implementations.

Energy efficiency:
RRAM has strong potential for very low-energy operation, while MRAM provides an established combination of speed, endurance and non-volatility.

Conclusion

MRAM: Best suited when fast access + high endurance + non-volatility are important.

RRAM: Attractive when high density + low power + non-volatility are priorities.

Overall, both technologies can complement conventional SRAM/DRAM/NAND in future hierarchical memory systems.